Solid State Physics — Week 8 Assignment Questions & Answers

Question 71

Find the energy level in sodium for which the probability of occupation at 300 K is 0.75. The Fermi energy of sodium is 3.13 eV. (Boltzmann constant$K_B = 1.38 \times 10^{-23}\ J/K$)

(a) $3.10\ eV$

(b) $2.28\ eV$

(c) $1.97\ eV$

(d) $1.12\ eV$

Accepted answer: (a)

Question 72

Find the temperature at which there is a 1% probability than an electron in a solid will have energy $0.5\ eV$ above the Fermi energy. (Boltzmann constant$K_B = 1.38 \times 10^{-23}\ J/K$)

(a) $1261.6\ K$

(b) $1322.7\ K$

(c) $1466.8\ K$

(d) $1547.6\ K$

Accepted answer: (a)

Question 73

The concentration of free electrons in sodium is$(2.5 \times 10^{28}\ /m^3)$. Calculate the Fermi energy at 0 K. (Given: Planck constant$(h = 6.64 \times 10^{-34}\ J-s)$)

(a) $1.23\ eV$

(b) $2.17\ eV$

(c) $3.13\ eV$

(d) $3.73\ eV$

Accepted answer: (c)

Question 74

Concentration of free electron density in sodium is$(2.5 \times 10^{28}/m^3)$. Calculate the Fermi velocity. [Given: Planck constant$(h = 6.64 \times 10^{-34}\ J-s)$]

(a) $1.12 \times 10^5\ m/s$

(b) $1.05 \times 10^6\ m/s$

(c) $2.17 \times 10^7\ m/s$

(d) $1.97 \times 10^9\ m/s$

Accepted answer: (b)

Question 75

The number of electrons per unit length of crystal is 0.5 electrons per angstrom. The total number of electrons, those are to be accommodate on a line of a one-dimensional crystal, is even number. Determine the Fermi energy of the system. [Given: Planck constant$(h = 6.62 \times 10^{-34}\ J - s)$]

(a) 2.35 eV

(b) 2.75 eV

(c) 3.25 eV

(d) 3.75 eV

Accepted answer: (a)

Question 76

Consider silver in the metallic with one free electron per atom. Calculate the Fermi energy at 0K. [Given: Atomic weight of Silver is 108, density of silver is$10.5\ gm/cc$, Planck constant$(h = 6.62 \times 10^{-34}\ J - s)$]

(a) 5.5 eV

(b) 4.5 eV

(c) 3.5 eV

(d) 2.5 eV

Accepted answer: (a)

Question 77

The atomic radius of sodium (BCC) is 1.86 Å. Calculate the Fermi energy of sodium at the temperature of 0K. Consider each sodium atom contribute one free electron. [Planck constant$(h = 6.62 \times 10^{-34}\, J - s)$]

(a) $3.12\ eV$

(b) $4.42\ eV$

(c) $5.12\ eV$

(d) $5.57\ eV$

Accepted answer: (a)

Question 78

Calculate the de-Broglie wavelength associated by an electron, in a hypothetical metal, having energy equal to the Fermi energy at 0 K. Consider numbers of free electrons per unit volume in the metal is n.

(a) $2\left(\frac{\pi}{3n}\right)^{\frac{1}{3}}$

(b) $2\left(\frac{3n}{\pi}\right)^{\frac{1}{3}}$

(c) $(3\pi^2 n)^{\frac{2}{3}}$

(d) $\left(\frac{3n}{\pi}\right)^{\frac{3}{2}}$

Accepted answer: (a)

Question 79

Silver is a monovalent metal having FCC crystal structure. If the atomic radius $r = 1.44\,\text{Å}$, calculate the Fermi energy at 0 K. Planck constant $(h = 6.62 \times 10^{-34}\ J - s)$.

(a) $2.17\ eV$

(b) $3.73\ eV$

(c) $4.23\ eV$

(d) $5.53\ eV$

Accepted answer: (d)

Question 80

The density of free electron states in a two-dimensional solid having dimension $(a \times a)$ varies with energy E as

(a) $E^{\frac{1}{2}}$

(b) $E^{\frac{-1}{2}}$

(c) E^{0}

(d) $E^{\frac{3}{2}}$

Accepted answer: (c)